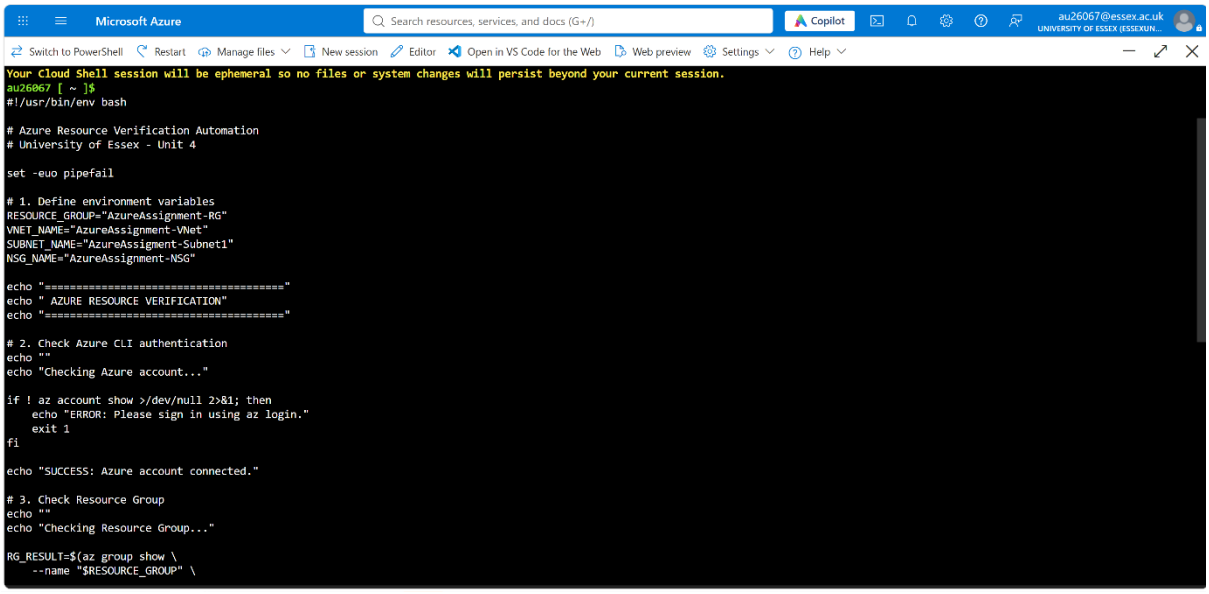
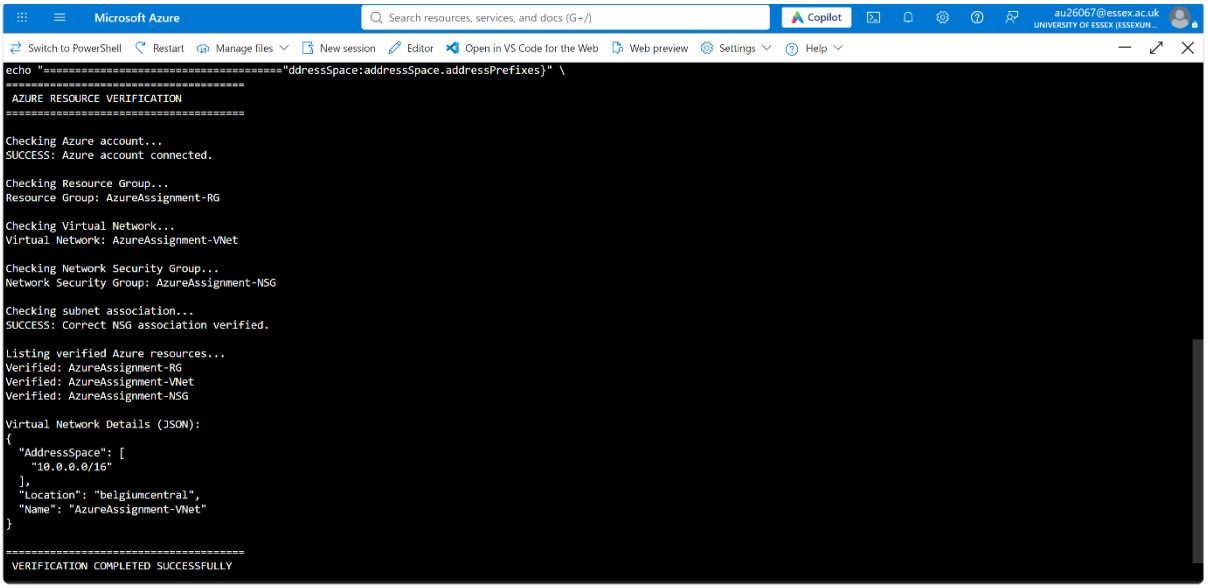
Formative Activity: Bash Script and Report

1. Automated Azure Provisioning & Verification Bash Script



Picture no.1



Picture no. 2

2. How Scripting Automation Enhances Cloud Operations

Using shell scripting to automate cloud operations converts a human-administrative process into automated, deterministic infrastructure-as-code (IaC). In large organisations' cloud services, e.g., Microsoft Azure, manually deploying virtual machines, virtual networks, firewall rules, etc. is extremely slow, prone to human errors, and poses a huge security risk. Shell scripting solves those issues by providing fast, consistent, idempotent, secure, cost-effective and easily auditable deployments of cloud infrastructure components.

3. Operational Speed and Lifecycle Agility

In enterprise cloud operations, the faster you can deploy, the faster you are able to respond to your business needs. Deploying a single instance via the Azure Portal by clicking through various blades, typing in subnet IP addresses, associating NSGs and configuring other properties can take a systems engineer 30-45 minutes to provision each instance. This simply isn't viable in any enterprise environment. Alternatively, provision an environment with a parameterised Bash script, which will create a Resource Group, a subnet, a Virtual Network, bind a Network Security Group (NSG) to the Subnet, deploy a VM, and then bootstrap a Package through the az vm run-command command in less than two minutes (Microsoft Learn, 2026).

Scripts can also easily be incorporated into a Continuous Integration and Continuous Deployment (CI/CD) pipeline to enable automated creation of environments for development, test, staging, and production.

4. Consistency and Elimination of Configuration Drift

Configuration drift caused by manual cloud operations leads to security issues and downtime. Shell scripts make infrastructure as code, replacing manual processes with defined variables like VNET_PREFIX="10.0.0.0/16" to enforce a consistent networking policy across all environments (Van Deun, 2025).

5. Idempotency, Error Handling, and Operational Resilience

Reliable cloud automation requires idempotency so that deployment scripts can be run multiple times without errors (Van Deun, 2025). Rigorous flags (set -euo pipefail) cause the script to exit in the event of a failure (Panovski, 2026), and conditional pre-flight checks ensure safe state transitions**.**

6. Automated Verification, Compliance, and Auditability

Manual cloud tasks introduce configuration drift, causing security vulnerabilities and outages. Shell scripts eliminate human error by codifying infrastructure into explicit variables like VNET_PREFIX="10.0.0.0/16", ensuring every environment strictly adheres to networking standards (Van Deun, 2025).

7. Analysis of Source Evidence Images (picture no. 1 & picture no. 2)

The implementation and execution of this cloud automation workflow are directly supported and validated by the provided source screenshots from the Microsoft Azure Cloud Shell session (au26067@essex.ac.uk - University of Essex):

- **Script Authoring (picture no. 1):** Captures the script structure inside the Azure Cloud Shell terminal editor under the student account au26067@essex.ac.uk. It incorporates strict error handling (set -euo pipefail), establishes resource variables (AzureAssignment-RG, AzureAssignment-VNet, AzureAssignment-NSG), and enforces an authentication gate via az account show.
- **Execution Telemetry (picture no.2):** Displays live terminal output upon execution, validating active Azure authentication, verifying NSG subnet binding (SUCCESS: Correct NSG association verified), and returning structured JSON telemetry confirming the 10.0.0.0/16 address space in region belgiumcentral. The UK location was initially selected but was not working due to a location-related issue, so belgiumcentral was chosen as an alternative to successfully complete the deployment and verification. The process concludes with VERIFICATION COMPLETED SUCCESSFULLY.

Reference

1. **DataCamp** (2025) How to Install Azure CLI: A Practical Guide for Installation, Authentication, and Usage. Available at:

https://www.datacamp.com/tutorial/how-to-install-azure-cli (Accessed: 11 November 2026).

2. **Microsoft Learn** (2026) Run scripts in a Linux VM in Azure using action Run Commands, Microsoft Azure Virtual Machines Documentation. Available at: https://learn.microsoft.com/en-us/azure/virtual-machines/linux/run-command (Accessed: 11 November 2026).
3. **Panovski, D.** (2026) Bash Strict Mode: set -euo pipefail Explained, Linuxize. Available at: https://linuxize.com/post/bash-strict-mode/ (Accessed: 11 November 2026).
4. **Van Deun, W.** (2025) Declarative vs Imperative Automation: Why It Matters for Infrastructure Success, OpsMill. Available at: https://opsmill.com/blog/declarative-vs-imperative-automation/ (Accessed: 11 November 2026).